those quantities, we simply do so. Of course, we can measure the system sloppily or crudely, thereby obtaining imprecise results or altering the system itself. But we don't have to; just by being careful, we can precisely measure everything there is to know about the system without altering it in any noticeable way. Classical mechanics offers a clear and unambiguous relationship between what we see and what the theory describes.

Quantum mechanics, for all its successes, offers no such thing. The enigma at the heart of quantum reality can be summed up in a simple motto: what we *see* when we look at the world seems to be fundamentally different from what actually *is*.

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Think about electrons, the elementary particles orbiting atomic nuclei, whose interactions are responsible for all of chemistry and hence almost everything interesting around you right now. As we did with the rock, we can ignore some of the electron's specific properties, like its spin and the fact that it has an electric field. (Really we could just stick with the rock as our example—rocks are quantum systems just as much as electrons are—but switching to a subatomic particle helps us remember that the features distinguishing quantum mechanics only become evident when we consider very tiny objects indeed.)

Unlike in classical mechanics, where the state of a system is described by its position and velocity, the nature of a quantum system is something a bit less concrete. Consider an electron in its natural habitat, orbiting the nucleus of an atom. You might think, from the word "orbit" as well as from the numerous cartoon depictions of atoms you have doubtless been exposed to over the years, that the orbit of an electron is more or less like the orbit of a planet in the solar system. The electron (so you might think) has a location, and a velocity, and as time passes it zips around the central nucleus in a circle or maybe an ellipse.

Quantum mechanics suggests something different. We can *measure* values of the location or velocity (though not at the same time), and if we are sufficiently careful and talented experimenters we will obtain some answer. But what we're seeing through such a measurement is not the actual, complete, unvarnished state of the electron. Indeed, the particular